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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	CodeBlack
Selected Domain	Domain 1 - AIVERSECITY
Selected SDG Track (1–17)	SDG 4,5,8,9,10
Project / Solution Name	Mustaqbal AI
Name (Leader)	Rehan Anjum
Email & Phone (Leader)	rehanjum9@gmail.com | +60 17-623 3902
University Affiliation(s)	UCSI University
Number of Team Members	4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

University students and recent graduates often struggle to identify suitable career paths because they lack accessible, personalized guidance connecting their academic qualifications, technical skills and project experience with real world employment opportunities. Traditional career exploration frequently depends on generic job listings, informal advice or time consuming research. Students may not understand how their current skills align with particular occupations, which competencies they need to develop or how changing job market trends affect their options. This challenge is especially relevant for young people entering an increasingly competitive and technology driven employment market. Students without professional networks or access to dedicated career counselling may face additional barriers to making informed decisions.

According to the International Labour Organization's Global Employment Trends for Youth 2024 approximately 20.4% of young people worldwide were not in employment, education or training in 2023. Although this statistic does not measure career guidance gaps directly, it illustrates the wider challenge of youth economic participation. Our project focuses on improving access to understandable, skills based career exploration. It supports SDG 8 by helping students identify potential career pathways and SDG 4 by encouraging the development of relevant employability skills.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Mustaqbal AI is a web based, AI assisted career exploration platform designed to help university students understand their skills, discover relevant career pathways and identify areas for professional development.

Students can create a profile manually or upload a PDF resume. The application uses GLM-5.3 through RunPod to extract structured information such as education, technical skills, projects and experience. Extracted information is presented in an editable review screen and users must confirm their details before proceeding. It provides ranked career

recommendations, interpretable skillalignment information and visibility into potential skill gaps. The prototype also incorporates cached job market trend information and SDG related career context.

The system uses React, TypeScript and Vite for the frontend, Python and FastAPI for backend APIs, SQLite with SQLAlchemy for persistence and Replit for cloud deployment. The RunPod-hosted GLM-5.3 API enables structured resume extraction, while the separate matching system provides explainable recommendations. The project integrates GenAI API usage, database architecture, full stack development and cloud deployment practices relevant to the hackathon's technical workshops.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend:
React, TypeScript, Vite and Tailwind CSS

Backend:
Python, FastAPI, Pydantic and REST APIs

Database:
SQLite with SQLAlchemy

AI Model and Provider:
GLM-5.3 accessed through RunPod's OpenAI compatible API for structured PDF resume extraction

Career Matching:
Hussain's deterministic, skills based matching algorithm with canonical skill normalization and explainable match information

Career and Trend Data:
Curated cached career catalogue, skills data, labour market trend snapshots and SDG mappings

Cloud Hosting:
Replit deployment

Version Control:
Git and GitHub

System Architecture:

Student → React Web Application → FastAPI Backend → PDF Text Extraction → RunPod GLM-5.3 → Structured CV Draft → Student Review and Confirmation → Skill Normalization → Career Matching Engine → Career Recommendations and SDG/Trend Information.

Career catalogue data and application information are supported by the database and locally available cached datasets.

Official SDC Sponsor-Provided GenAI API or Cloud Platform:

RunPod is used to access GLM-5.3 for AI-based CV information extraction

Used Replit for Deployment and for CV extraction timeout investigation, FastAPI improvements, diagnostics, implementation and automated testing

Mustaqbal AI uses OpenRouter to access Z.ai GLM-5.3 FlashX for AI assisted CV information extraction. Extracted information is presented as an editable draft and must be reviewed and confirmed by the student. Career recommendations are generated using an explainable weighted matching algorithm, with optional semantic skill similarity powered by a RunPod Serverless endpoint. Replit hosts the web application.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our team followed a student centred design approach focused on accessibility, practical usefulness and technical feasibility.

We identified the difficulty students experience when translating academic knowledge and technical skills into realistic career options. We then designed a straightforward workflow allowing users either to enter profile information manually or upload a resume.

The interface provides editable CV extraction results, transparent skill selection and explicit confirmation before generating recommendations. This reduces the risk of relying on incorrect AI generated information.

The development process separated responsibilities across frontend development, backend integration, career matching and data management. Modular APIs and a curated catalogue were used to keep the system testable and explainable.

The application was deployed using Replit, with backend regression tests, frontend compilation, lint checks and CV extraction testing supporting technical validation. A safe fallback is available when the external AI extraction service is unavailable.

For Phase 2, we aim to conduct structured usability testing, evaluate recommendation relevance, improve recognition of technical skills and measure whether the application reduces the time required for students to explore suitable career pathways.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Primary: SDG Target 8.6 – Reduce the proportion of youth not in employment, education or training. Supporting: SDG Target 4.4 – Increase the number of youth and adults with relevant skills for employment, decent jobs and entrepreneurship.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	International Labour Organization reported that approximately 20.4% of young people aged 15–24 worldwide were not in employment, education or training in 2023. This is a global contextual baseline, not a measured impact of Mustaqbal AI. For the platform specific baseline we propose measuring the average time required for students to identify three relevant career pathways and their associated skill gaps without Mustaqbal AI, using an initial pilot group of 20 students. The baseline time will be recorded before evaluating the application.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Percentage reduction in average career exploration task completion time compared with the manual research baseline. Secondary measurements: Average task completion time. Percentage of participants able to identify three relevant career pathways. Student reported usefulness of career recommendations.

Metric Category	Your Response
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Target a 20% reduction in average career-exploration task completion time compared with the measured manual baseline during a 20 student pilot. This is a proposed Phase 2 target, not an achieved result. Broader employment outcomes are outside the prototype's evaluation scope.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

No

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

Yes

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React	Code / frontend library	MIT
FastAPI	Code / backend framework	MIT
Vite	Code / build framework	MIT
SQLAlchemy	Code / ORM	MIT
Pydantic	Code / data validation	MIT
GLM-5.3 via RunPod	Third-party AI model and API	Applicable model/provider terms
Curated career and labour-market data	Dataset / reference data	Verify each original source and reuse terms
Other installed dependencies	Third-party code	Declare applicable licenses from dependency inventory
Deepseek Glm 5.3 flashx	Model using API	

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

No

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Rehan Anjum

25/10/2026